

Module 01: Systems in High School

Learning Objectives

1. Define **Systems** within the context of High School.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of **Systems**.

Introduction

This module explores **Systems**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

Applications

In High School, we see **Systems** manifest in: * **Specific Example 1:** A smartphone's operating system is an Active Inference system where the CPU, RAM, and battery form internal states,

the apps and user interactions are external states, and the touchscreen plus network interfaces form the Markov blanket -- the OS continuously predicts resource demands (how much RAM the next app needs) and adjusts allocation to minimize the surprise of lag or crashes. *

Specific Example 2: A smart home ecosystem (thermostat, lights, door locks, voice assistant) operates as a distributed Active Inference system: each device maintains its own generative model of expected conditions (temperature schedules, occupancy patterns), they communicate through a shared network (Markov blanket), and the system collectively minimizes household free energy by predicting and preemptively adjusting to the residents' routines.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.